



Національний технічний університет України  
«Київський політехнічний інститут імені Ігоря Сікорського»

Факультет інформатики та обчислювальної техніки  
Кафедра обчислювальної техніки

# Вступ до операційної системи Linux. Курсова робота

## Завдання №4

«Вивчення CI/CD інструментів»

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Перевірив: *Нікольський С. С.*

**Тема:** «Вивчення CI/CD інструментів».

**Мета:** навчитись користуванню інструментів CI/CD інструментів на основі Jenkins.

## Практична частина

### Робоча директорія:

```
1 PS C:\Users\artem\Desktop\project> tree /f /a
2 Folder PATH listing
3 Volume serial number is 1A5B-F8AF
4 C:.\
5 |   .gitignore
6 |   CMakeLists.txt
7 |
8 +---src
9 |       sort.cpp
10 |       sort.h
11 |       stringProcessor.cpp
12 |       stringProcessor.h
13 |
14 \---tests
15       smoke_test.cpp
```

### Програмний код src/sort.cpp:

```
1 #include "sort.h"
2
3 namespace algo {
4
5 void insertion_sort(std::vector<int>& a) {
6     for (size_t i = 1; i < a.size(); ++i) {
7         int key = a[i];
8         size_t j = i;
9         while (j > 0 && a[j - 1] > key) {
10             a[j] = a[j - 1];
11             --j;
12         }
13         a[j] = key;
14     }
15 }
16
17 bool is_sorted_non_decreasing(const std::vector<int>& a) {
18     for (size_t i = 1; i < a.size(); ++i) {
19         if (a[i - 1] > a[i]) return false;
20     }
21     return true;
22 }
23
24 } // namespace algo
```

## Програмний код src/sort.h:

```
1 #pragma once
2 #include <vector>
3
4 namespace algo {
5     // Стабільне просте сортування (in-place) - для прикладу
6     void insertion_sort(std::vector<int>& a);
7
8     // Перевірка відсортованості за нестрогим зростанням
9     bool is_sorted_non_decreasing(const std::vector<int>& a);
10 }
```

## Програмний код src/stringProcessor.cpp:

```
1 #include "stringProcessor.h"
2
3 namespace {
4     inline bool is_space(char c) {
5         return c == ' ' || c == '\t' || c == '\n' || c == '\r';
6     }
7 }
8
9 namespace sp {
10
11 std::string trim(const std::string& s) {
12     if (s.empty()) return s;
13     size_t i = 0, j = s.size();
14     while (i < j && is_space(s[i])) ++i;
15     while (j > i && is_space(s[j - 1])) --j;
16     return s.substr(i, j - i);
17 }
18
19 std::string to_lower(std::string s) {
20     for (auto& c : s) {
21         if (c >= 'A' && c <= 'Z') c = char(c - 'A' + 'a');
22     }
23     return s;
24 }
25
26 bool is_palindrome(const std::string& s) {
27     // ігноруємо пробіли, нечутливість до регістру
28     std::string t;
29     t.reserve(s.size());
30     for (char c : s) {
31         if (!is_space(c)) t.push_back(c);
32     }
33     t = to_lower(t);
34     size_t i = 0, j = t.size();
35     if (j == 0) return true;
36     --j;
37     while (i < j) {
38         if (t[i] != t[j]) return false;
39         ++i; --j;
40     }
41     return true;
42 }
```

```

40     }
41     return true;
42 }
43
44 } // namespace sp

```

### Програмний код src/stringProcessor.h:

```

1  #pragma once
2  #include <string>
3  #include <algorithm>
4
5  namespace sp {
6      // Трим пробіли з обох боків
7      std::string trim(const std::string& s);
8
9      // До нижнього регістру (ASCII)
10     std::string to_lower(std::string s);
11
12     // Чи паліндром (без урахування регістру і пробілів)
13     bool is_palindrome(const std::string& s);
14 }

```

### Конфігурація + генерація Solution (.sln):

```

1  PS C:\Users\artem\Desktop\project> cmake -S . -B build -G "Visual Studio 17 2022"
2  -- Selecting Windows SDK version 10.0.26100.0 to target Windows 6.2.9200.
3  -- The CXX compiler identification is MSVC 19.44.35219.0
4  -- Detecting CXX compiler ABI info
5  -- Detecting CXX compiler ABI info - done
6  -- Check for working CXX compiler: C:/Program Files (x86)/Microsoft Visual
   ↳ Studio/2022/BuildTools/VC/Tools/MSVC/14.44.35207/bin/Hostx64/x64/cl.exe -
   ↳ skipped
7  -- Detecting CXX compile features
8  -- Detecting CXX compile features - done
9  CMake Warning (dev) at
   ↳ C:/Strawberry/c/share/cmake-3.29/Modules/FetchContent.cmake:1352 (message):
10     The DOWNLOAD_EXTRACT_TIMESTAMP option was not given and policy CMP0135 is
11     not set. The policy's OLD behavior will be used. When using a URL
12     download, the timestamps of extracted files should preferably be that of
13     the time of extraction, otherwise code that depends on the extracted
14     contents might not be rebuilt if the URL changes. The OLD behavior
15     preserves the timestamps from the archive instead, but this is usually not
16     what you want. Update your project to the NEW behavior or specify the
17     DOWNLOAD_EXTRACT_TIMESTAMP option with a value of true to avoid this
18     robustness issue.
19  Call Stack (most recent call first):
20     CMakeLists.txt:17 (FetchContent_Declare)
21  This warning is for project developers. Use -Wno-dev to suppress it.
22
23  -- The C compiler identification is MSVC 19.44.35219.0
24  -- Detecting C compiler ABI info
25  -- Detecting C compiler ABI info - done

```

```

26 -- Check for working C compiler: C:/Program Files (x86)/Microsoft Visual
   ↳ Studio/2022/BuildTools/VC/Tools/MSVC/14.44.35207/bin/Hostx64/x64/cl.exe -
   ↳ skipped
27 -- Detecting C compile features
28 -- Detecting C compile features - done
29 -- Found Python3: C:/Users/artem/AppData/Local/Programs/Python/Python313/python.exe
   ↳ (found version "3.13.6") found components: Interpreter
30 -- Performing Test CMAKE_HAVE_LIBC_PTHREAD
31 -- Performing Test CMAKE_HAVE_LIBC_PTHREAD - Failed
32 -- Looking for pthread_create in pthreads
33 -- Looking for pthread_create in pthreads - not found
34 -- Looking for pthread_create in pthread
35 -- Looking for pthread_create in pthread - not found
36 -- Found Threads: TRUE
37 -- Configuring done (22.1s)
38 -- Generating done (0.2s)
39 -- Build files have been written to: C:/Users/artem/Desktop/project/build

```

### 36ipka (Debug):

```

1 PS C:\Users\artem\Desktop\project> cmake --build build --config Debug
2 MSBuild version 17.14.23+b0019275e for .NET Framework
3
4 1>Checking Build System
5 Building Custom Rule
   ↳ C:/Users/artem/Desktop/project/build/_deps/googletest-src/googlemock/
6 CMakeLists.txt
7 gtest-all.cc
8 gmock-all.cc
9 Generating Code...
10 gmock.vcxproj -> C:\Users\artem\Desktop\project\build\lib\Debug\gmock.lib
11 Building Custom Rule
   ↳ C:/Users/artem/Desktop/project/build/_deps/googletest-src/googlemock/
12 CMakeLists.txt
13 gtest-all.cc
14 gmock-all.cc
15 gmock_main.cc
16 Generating Code...
17 gmock_main.vcxproj ->
   ↳ C:\Users\artem\Desktop\project\build\lib\Debug\gmock_main.lib
18 Building Custom Rule
   ↳ C:/Users/artem/Desktop/project/build/_deps/googletest-src/googletest/
19 CMakeLists.txt
20 gtest-all.cc
21 gtest.vcxproj -> C:\Users\artem\Desktop\project\build\lib\Debug\gtest.lib
22 Building Custom Rule
   ↳ C:/Users/artem/Desktop/project/build/_deps/googletest-src/googletest/
23 CMakeLists.txt
24 gtest_main.cc
25 gtest_main.vcxproj ->
   ↳ C:\Users\artem\Desktop\project\build\lib\Debug\gtest_main.lib
26 Building Custom Rule C:/Users/artem/Desktop/project/CMakeLists.txt
27 stringProcessor.cpp
28 sort.cpp

```

```

29 Generating Code...
30 my_lib.vcxproj -> C:\Users\artem\Desktop\project\build\Debug\my_lib.lib
31 Building Custom Rule C:/Users/artem/Desktop/project/CMakeLists.txt
32 smoke_test.cpp
33 unit_tests.vcxproj -> C:\Users\artem\Desktop\project\build\Debug\unit_tests.exe
34 Building Custom Rule C:/Users/artem/Desktop/project/CMakeLists.txt

```

### Запуск тестів з генерацією XML-звіту:

```

1 PS C:\Users\artem\Desktop\project> .\build\Debug\unit_tests.exe
  ↳ --gtest_output=xml:build\test_report.xml
2 Running main() from
  ↳ C:\Users\artem\Desktop\project\build\_deps\googletest-src\googletest\src
3 gtest_main.cc
4 [=====] Running 3 tests from 2 test suites.
5 [-----] Global test environment set-up.
6 [-----] 2 tests from StringProcessor
7 [ RUN      ] StringProcessor.TrimAndLower
8 [          OK ] StringProcessor.TrimAndLower (0 ms)
9 [ RUN      ] StringProcessor.Palindrome
10 [          OK ] StringProcessor.Palindrome (0 ms)
11 [-----] 2 tests from StringProcessor (1 ms total)
12
13 [-----] 1 test from Sort
14 [ RUN      ] Sort.InsertionSortWorks
15 [          OK ] Sort.InsertionSortWorks (0 ms)
16 [-----] 1 test from Sort (0 ms total)
17
18 [-----] Global test environment tear-down
19 [=====] 3 tests from 2 test suites ran. (4 ms total)
20 [ PASSED   ] 3 tests.

```

### Робоча директорія після виконаних кроків конфігурації, генерації та збірки:

```

1 PS C:\Users\artem\Desktop\project> tree /f /a
2 Folder PATH listing
3 Volume serial number is 1A5B-F8AF
4 C:.\
5 |   .consolestory
6 |   .gitignore
7 |   CMakeLists.txt
8 |   t.txt
9 |
10 +---build
11 |   |   ALL_BUILD.vcxproj
12 |   |   ALL_BUILD.vcxproj.filters
13 |   |   CMakeCache.txt
14 |   |   cmake_install.cmake
15 |   |   CTestTestfile.cmake
16 |   |   INSTALL.vcxproj
17 |   |   INSTALL.vcxproj.filters
18 |   |   MyGTestProj.sln
19 |   |   my_lib.vcxproj

```

```

20 | | my_lib.vcxproj.filters
21 | | RUN_TESTS.vcxproj
22 | | RUN_TESTS.vcxproj.filters
23 | | test_report.xml
24 | | unit_tests.vcxproj
25 | | unit_tests.vcxproj.filters
26 | | unit_tests[1]_include.cmake
27 | | unit_tests[1]_tests.cmake
28 | | ZERO_CHECK.vcxproj
29 | | ZERO_CHECK.vcxproj.filters
30 | |
31 | +---CMakeFiles
32 | | | cmake.check_cache
33 | | | CMakeConfigureLog.yaml
34 | | | generate.stamp
35 | | | generate.stamp.depend
36 | | | generate.stamp.list
37 | | | TargetDirectories.txt
38 | | |
39 | | +---221e8778a9cacabaa0e7dca956df043d
40 | | | INSTALL_force.rule
41 | | | RUN_TESTS_force.rule
42 | | |
43 | | +---3.29.2
44 | | | CMakeCCompiler.cmake
45 | | | CMakeCXXCompiler.cmake
46 | | | CMakeDetermineCompilerABI_C.bin
47 | | | CMakeDetermineCompilerABI_CXX.bin
48 | | | CMakeRCCompiler.cmake
49 | | | CMakeSystem.cmake
50 | | | VCTargetsPath.txt
51 | | | VCTargetsPath.vcxproj
52 | | |
53 | | | +---CompilerIdC
54 | | | | CMakeCCompilerId.c
55 | | | | CompilerIdC.exe
56 | | | | CompilerIdC.vcxproj
57 | | | |
58 | | | | +---Debug
59 | | | | | CMakeCCompilerId.obj
60 | | | | | CompilerIdC.exe.recipe
61 | | | | |
62 | | | | \---CompilerIdC.tlog
63 | | | | | CL.command.1.tlog
64 | | | | | CL.items.tlog
65 | | | | | CL.read.1.tlog
66 | | | | | CL.write.1.tlog
67 | | | | | CompilerIdC.lastbuildstate
68 | | | | | link.command.1.tlog
69 | | | | | link.read.1.tlog
70 | | | | | link.secondary.1.tlog
71 | | | | | link.write.1.tlog
72 | | | |
73 | | | \---tmp

```

```

74 | | | +---CompilerIdCXX
75 | | | | CMakeCXXCompilerId.cpp
76 | | | | CompilerIdCXX.exe
77 | | | | CompilerIdCXX.vcxproj
78 | | | |
79 | | | +---Debug
80 | | | | CMakeCXXCompilerId.obj
81 | | | | CompilerIdCXX.exe.recipe
82 | | | |
83 | | | \---CompilerIdCXX.tlog
84 | | | | CL.command.1.tlog
85 | | | | CL.items.tlog
86 | | | | CL.read.1.tlog
87 | | | | CL.write.1.tlog
88 | | | | CompilerIdCXX.lastbuildstate
89 | | | | link.command.1.tlog
90 | | | | link.read.1.tlog
91 | | | | link.secondary.1.tlog
92 | | | | link.write.1.tlog
93 | | | |
94 | | | \---tmp
95 | | | +---VCTargetsPath
96 | | | \---x64
97 | | | \---Debug
98 | | | | VCTargetsPath.recipe
99 | | | |
100 | | | \---VCTargetsPath.tlog
101 | | | | VCTargetsPath.lastbuildstate
102 | | | |
103 | | | \---x64
104 | | | \---Debug
105 | | +---ba30693984164defdda498b2fc47b999
106 | | | generate.stamp.rule
107 | | | INSTALL_force.rule
108 | | | RUN_TESTS_force.rule
109 | | |
110 | | +---CMakeScratch
111 | | +---e1016b5e72a78186d6d2c7373e841be2
112 | | | INSTALL_force.rule
113 | | | RUN_TESTS_force.rule
114 | | |
115 | | +---e5097ab0b0bc71e539ee0b41c04d6012
116 | | | INSTALL_force.rule
117 | | | RUN_TESTS_force.rule
118 | | |
119 | | \---pkgRedirects
120 | +---Debug
121 | | my_lib.lib
122 | | my_lib.pdb
123 | | unit_tests.exe
124 | | unit_tests.pdb
125 | |
126 | +---lib
127 | | \---Debug

```



```

128 | | gmock.lib
129 | | gmock.pdb
130 | | gmockpdb_debug_postfix-NOTFOUND.pdb
131 | | gmock_main.lib
132 | | gmock_main.pdb
133 | | gmock_mainpdb_debug_postfix-NOTFOUND.pdb
134 | | gtest.lib
135 | | gtest.pdb
136 | | gtestpdb_debug_postfix-NOTFOUND.pdb
137 | | gtest_main.lib
138 | | gtest_main.pdb
139 | | gtest_mainpdb_debug_postfix-NOTFOUND.pdb
140 | |
141 | +---my_lib.dir
142 | | \---Debug
143 | | | my_lib.lib.recipe
144 | | | sort.obj
145 | | | stringProcessor.obj
146 | | |
147 | | \---my_lib.tlog
148 | | CL.command.1.tlog
149 | | Cl.items.tlog
150 | | CL.read.1.tlog
151 | | CL.write.1.tlog
152 | | CustomBuild.command.1.tlog
153 | | CustomBuild.read.1.tlog
154 | | CustomBuild.write.1.tlog
155 | | Lib-link.read.1.tlog
156 | | Lib-link.write.1.tlog
157 | | Lib.command.1.tlog
158 | | my_lib.lastbuildstate
159 | |
160 | +---unit_tests.dir
161 | | \---Debug
162 | | | smoke_test.obj
163 | | | unit_tests.exe.recipe
164 | | | unit_tests.ilc
165 | | | vc143.pdb
166 | | |
167 | | \---unit_tests.tlog
168 | | CL.command.1.tlog
169 | | Cl.items.tlog
170 | | CL.read.1.tlog
171 | | CL.write.1.tlog
172 | | CustomBuild.command.1.tlog
173 | | CustomBuild.read.1.tlog
174 | | CustomBuild.write.1.tlog
175 | | link.command.1.tlog
176 | | link.read.1.tlog
177 | | link.secondary.1.tlog
178 | | link.write.1.tlog
179 | | unit_tests.lastbuildstate
180 | |
181 | +---x64

```

```

182 | | \---Debug
183 | |     +---ALL_BUILD
184 | |         | | ALL_BUILD.recipe
185 | |         | |
186 | |         | | \---ALL_BUILD.tlog
187 | |             | | ALL_BUILD.lastbuildstate
188 | |             | | CustomBuild.command.1.tlog
189 | |             | | CustomBuild.read.1.tlog
190 | |             | | CustomBuild.write.1.tlog
191 | |             | |
192 | |         | | \---ZERO_CHECK
193 | |             | | ZERO_CHECK.recipe
194 | |             | |
195 | |             | | \---ZERO_CHECK.tlog
196 | |                 | | CustomBuild.command.1.tlog
197 | |                 | | CustomBuild.read.1.tlog
198 | |                 | | CustomBuild.write.1.tlog
199 | |                 | | ZERO_CHECK.lastbuildstate
200 | |
201 | | \---_deps
202 | |     +---googletest-build
203 | |         | | ALL_BUILD.vcxproj
204 | |         | | ALL_BUILD.vcxproj.filters
205 | |         | | cmake_install.cmake
206 | |         | | CTestTestfile.cmake
207 | |         | | googletest-distribution.sln
208 | |         | | INSTALL.vcxproj
209 | |         | | INSTALL.vcxproj.filters
210 | |         | | RUN_TESTS.vcxproj
211 | |         | | RUN_TESTS.vcxproj.filters
212 | |         | |
213 | |         | | +---CMakeFiles
214 | |             | | generate.stamp
215 | |             | | generate.stamp.depend
216 | |             | |
217 | |         | | +---googlemock
218 | |             | | ALL_BUILD.vcxproj
219 | |             | | ALL_BUILD.vcxproj.filters
220 | |             | | cmake_install.cmake
221 | |             | | CTestTestfile.cmake
222 | |             | | gmock.sln
223 | |             | | gmock.vcxproj
224 | |             | | gmock.vcxproj.filters
225 | |             | | gmock_main.vcxproj
226 | |             | | gmock_main.vcxproj.filters
227 | |             | | INSTALL.vcxproj
228 | |             | | INSTALL.vcxproj.filters
229 | |             | | RUN_TESTS.vcxproj
230 | |             | | RUN_TESTS.vcxproj.filters
231 | |             | |
232 | |         | | +---CMakeFiles
233 | |             | | generate.stamp
234 | |             | | generate.stamp.depend
235 | |

```

```

236 | | | +---gmock.dir
237 | | | | \---Debug
238 | | | | | gmock-all.obj
239 | | | | | gmock.lib.recipe
240 | | | | | gtest-all.obj
241 | | | | |
242 | | | | \---gmock.tlog
243 | | | | | CL.command.1.tlog
244 | | | | | CL.items.tlog
245 | | | | | CL.read.1.tlog
246 | | | | | CL.write.1.tlog
247 | | | | | CustomBuild.command.1.tlog
248 | | | | | CustomBuild.read.1.tlog
249 | | | | | CustomBuild.write.1.tlog
250 | | | | | gmock.lastbuildstate
251 | | | | | Lib-link.read.1.tlog
252 | | | | | Lib-link.write.1.tlog
253 | | | | | Lib.command.1.tlog
254 | | | |
255 | | | \---gmock_main.dir
256 | | | | \---Debug
257 | | | | | gmock-all.obj
258 | | | | | gmock_main.lib.recipe
259 | | | | | gmock_main.obj
260 | | | | | gtest-all.obj
261 | | | | |
262 | | | | \---gmock_main.tlog
263 | | | | | CL.command.1.tlog
264 | | | | | CL.items.tlog
265 | | | | | CL.read.1.tlog
266 | | | | | CL.write.1.tlog
267 | | | | | CustomBuild.command.1.tlog
268 | | | | | CustomBuild.read.1.tlog
269 | | | | | CustomBuild.write.1.tlog
270 | | | | | gmock_main.lastbuildstate
271 | | | | | Lib-link.read.1.tlog
272 | | | | | Lib-link.write.1.tlog
273 | | | | | Lib.command.1.tlog
274 | | | |
275 | | | \---googletest
276 | | | | ALL_BUILD.vcxproj
277 | | | | ALL_BUILD.vcxproj.filters
278 | | | | cmake_install.cmake
279 | | | | CTestTestfile.cmake
280 | | | | gtest.sln
281 | | | | gtest.vcxproj
282 | | | | gtest.vcxproj.filters
283 | | | | gtest_main.vcxproj
284 | | | | gtest_main.vcxproj.filters
285 | | | | INSTALL.vcxproj
286 | | | | INSTALL.vcxproj.filters
287 | | | | RUN_TESTS.vcxproj
288 | | | | RUN_TESTS.vcxproj.filters
289 | | | |

```

```

290 | | +---CMakeFiles
291 | | | generate.stamp
292 | | | generate.stamp.depend
293 | | |
294 | | | \---Export
295 | | | | \---0c08b8e77dd885bfe55a19a9659d9fc1
296 | | | | GTestTargets-debug.cmake
297 | | | | GTestTargets-minsizerel.cmake
298 | | | | GTestTargets-release.cmake
299 | | | | GTestTargets-relwithdebinfo.cmake
300 | | | | GTestTargets.cmake
301 | | |
302 | | +---generated
303 | | | gmock.pc
304 | | | gmock_main.pc
305 | | | gtest.pc
306 | | | GTestConfig.cmake
307 | | | GTestConfigVersion.cmake
308 | | | gtest_main.pc
309 | | |
310 | | +---gtest.dir
311 | | | \---Debug
312 | | | | gtest-all.obj
313 | | | | gtest.lib.recipe
314 | | | |
315 | | | | \---gtest.tlog
316 | | | | CL.command.1.tlog
317 | | | | CL.items.tlog
318 | | | | CL.read.1.tlog
319 | | | | CL.write.1.tlog
320 | | | | CustomBuild.command.1.tlog
321 | | | | CustomBuild.read.1.tlog
322 | | | | CustomBuild.write.1.tlog
323 | | | | gtest.lastbuildstate
324 | | | | Lib-link.read.1.tlog
325 | | | | Lib-link.write.1.tlog
326 | | | | Lib.command.1.tlog
327 | | |
328 | | \---gtest_main.dir
329 | | | \---Debug
330 | | | | gtest_main.lib.recipe
331 | | | | gtest_main.obj
332 | | | |
333 | | | | \---gtest_main.tlog
334 | | | | CL.command.1.tlog
335 | | | | CL.items.tlog
336 | | | | CL.read.1.tlog
337 | | | | CL.write.1.tlog
338 | | | | CustomBuild.command.1.tlog
339 | | | | CustomBuild.read.1.tlog
340 | | | | CustomBuild.write.1.tlog
341 | | | | gtest_main.lastbuildstate
342 | | | | Lib-link.read.1.tlog
343 | | | | Lib-link.write.1.tlog

```

```

344 | | | | | Lib.command.1.tlog
345 | | | | |
346 | | | | | +---googletest-src
347 | | | | | | | .clang-format
348 | | | | | | | .gitignore
349 | | | | | | | BUILD.bazel
350 | | | | | | | CMakeLists.txt
351 | | | | | | | CONTRIBUTING.md
352 | | | | | | | CONTRIBUTORS
353 | | | | | | | googletest_deps.bzl
354 | | | | | | | LICENSE
355 | | | | | | | README.md
356 | | | | | | | WORKSPACE
357 | | | | | | |
358 | | | | | | +---.github
359 | | | | | | | | +---ISSUE_TEMPLATE
360 | | | | | | | | | 00-bug_report.yml
361 | | | | | | | | | 10-feature_request.yml
362 | | | | | | | | | config.yml
363 | | | | | | | | |
364 | | | | | | | | \---workflows
365 | | | | | | | | | gtest-ci.yml
366 | | | | | | | | |
367 | | | | | | | +---ci
368 | | | | | | | | linux-presubmit.sh
369 | | | | | | | | macos-presubmit.sh
370 | | | | | | | | windows-presubmit.bat
371 | | | | | | | |
372 | | | | | | | +---docs
373 | | | | | | | | advanced.md
374 | | | | | | | | community_created_documentation.md
375 | | | | | | | | faq.md
376 | | | | | | | | gmock_cheat_sheet.md
377 | | | | | | | | gmock_cook_book.md
378 | | | | | | | | gmock_faq.md
379 | | | | | | | | gmock_for_dummies.md
380 | | | | | | | | index.md
381 | | | | | | | | pkgconfig.md
382 | | | | | | | | platforms.md
383 | | | | | | | | primer.md
384 | | | | | | | | quickstart-bazel.md
385 | | | | | | | | quickstart-cmake.md
386 | | | | | | | | samples.md
387 | | | | | | | | _config.yml
388 | | | | | | | |
389 | | | | | | | +---assets
390 | | | | | | | | \---css
391 | | | | | | | | | style.scss
392 | | | | | | | | |
393 | | | | | | | +---reference
394 | | | | | | | | actions.md
395 | | | | | | | | assertions.md
396 | | | | | | | | matchers.md
397 | | | | | | | | mocking.md

```

```

398 | | | | testing.md
399 | | | |
400 | | | +---_data
401 | | | | navigation.yml
402 | | | |
403 | | | +---_layouts
404 | | | | default.html
405 | | | |
406 | | | \---_sass
407 | | | | main.scss
408 | | | |
409 | | | +---googlemock
410 | | | | CMakeLists.txt
411 | | | | README.md
412 | | | |
413 | | | +---cmake
414 | | | | gmock.pc.in
415 | | | | gmock_main.pc.in
416 | | | |
417 | | | +---docs
418 | | | | README.md
419 | | | |
420 | | | +---include
421 | | | | \---gmock
422 | | | | | gmock-actions.h
423 | | | | | gmock-cardinalities.h
424 | | | | | gmock-function-mocker.h
425 | | | | | gmock-matchers.h
426 | | | | | gmock-more-actions.h
427 | | | | | gmock-more-matchers.h
428 | | | | | gmock-nice-strict.h
429 | | | | | gmock-spec-builders.h
430 | | | | | gmock.h
431 | | | | |
432 | | | | \---internal
433 | | | | | gmock-internal-utils.h
434 | | | | | gmock-port.h
435 | | | | | gmock-pp.h
436 | | | | |
437 | | | | \---custom
438 | | | | | gmock-generated-actions.h
439 | | | | | gmock-matchers.h
440 | | | | | gmock-port.h
441 | | | | | README.md
442 | | | |
443 | | | +---src
444 | | | | gmock-all.cc
445 | | | | gmock-cardinalities.cc
446 | | | | gmock-internal-utils.cc
447 | | | | gmock-matchers.cc
448 | | | | gmock-spec-builders.cc
449 | | | | gmock.cc
450 | | | | gmock_main.cc
451 | | | |

```

```

452 | | | \---test
453 | | |     BUILD.bazel
454 | | |     gmock-actions_test.cc
455 | | |     gmock-cardinalities_test.cc
456 | | |     gmock-function-mocker_test.cc
457 | | |     gmock-internal-utils_test.cc
458 | | |     gmock-matchers-arithmetic_test.cc
459 | | |     gmock-matchers-comparisons_test.cc
460 | | |     gmock-matchers-containers_test.cc
461 | | |     gmock-matchers-misc_test.cc
462 | | |     gmock-matchers_test.h
463 | | |     gmock-more-actions_test.cc
464 | | |     gmock-nice-strict_test.cc
465 | | |     gmock-port_test.cc
466 | | |     gmock-pp-string_test.cc
467 | | |     gmock-pp_test.cc
468 | | |     gmock-spec-builders_test.cc
469 | | |     gmock_all_test.cc
470 | | |     gmock_ex_test.cc
471 | | |     gmock_leak_test.py
472 | | |     gmock_leak_test_.cc
473 | | |     gmock_link2_test.cc
474 | | |     gmock_link_test.cc
475 | | |     gmock_link_test.h
476 | | |     gmock_output_test.py
477 | | |     gmock_output_test_.cc
478 | | |     gmock_output_test_golden.txt
479 | | |     gmock_stress_test.cc
480 | | |     gmock_test.cc
481 | | |     gmock_test_utils.py
482 | | |
483 | | | \---googletest
484 | | |     CMakeLists.txt
485 | | |     README.md
486 | | |
487 | | | +---cmake
488 | | | |     Config.cmake.in
489 | | | |     gtest.pc.in
490 | | | |     gtest_main.pc.in
491 | | | |     internal_utils.cmake
492 | | | |     libgtest.la.in
493 | | |
494 | | | +---docs
495 | | | |     README.md
496 | | |
497 | | | +---include
498 | | | |     \---gtest
499 | | | | |     gtest-assertion-result.h
500 | | | | |     gtest-death-test.h
501 | | | | |     gtest-matchers.h
502 | | | | |     gtest-message.h
503 | | | | |     gtest-param-test.h
504 | | | | |     gtest-printers.h
505 | | | | |     gtest-spi.h

```

```

506 | | | | gtest-test-part.h
507 | | | | gtest-typed-test.h
508 | | | | gtest.h
509 | | | | gtest_pred_impl.h
510 | | | | gtest_prod.h
511 | | | |
512 | | | | \---internal
513 | | | | | gtest-death-test-internal.h
514 | | | | | gtest-filepath.h
515 | | | | | gtest-internal.h
516 | | | | | gtest-param-util.h
517 | | | | | gtest-port-arch.h
518 | | | | | gtest-port.h
519 | | | | | gtest-string.h
520 | | | | | gtest-type-util.h
521 | | | | |
522 | | | | \---custom
523 | | | | | gtest-port.h
524 | | | | | gtest-printers.h
525 | | | | | gtest.h
526 | | | | | README.md
527 | | | |
528 | | | +---samples
529 | | | | prime_tables.h
530 | | | | sample1.cc
531 | | | | sample1.h
532 | | | | sample10_unittest.cc
533 | | | | sample1_unittest.cc
534 | | | | sample2.cc
535 | | | | sample2.h
536 | | | | sample2_unittest.cc
537 | | | | sample3-inl.h
538 | | | | sample3_unittest.cc
539 | | | | sample4.cc
540 | | | | sample4.h
541 | | | | sample4_unittest.cc
542 | | | | sample5_unittest.cc
543 | | | | sample6_unittest.cc
544 | | | | sample7_unittest.cc
545 | | | | sample8_unittest.cc
546 | | | | sample9_unittest.cc
547 | | | |
548 | | | +---src
549 | | | | gtest-all.cc
550 | | | | gtest-assertion-result.cc
551 | | | | gtest-death-test.cc
552 | | | | gtest-filepath.cc
553 | | | | gtest-internal-inl.h
554 | | | | gtest-matchers.cc
555 | | | | gtest-port.cc
556 | | | | gtest-printers.cc
557 | | | | gtest-test-part.cc
558 | | | | gtest-typed-test.cc
559 | | | | gtest.cc

```



```

560 | | | gtest_main.cc
561 | | |
562 | | | \---test
563 | | | BUILD.bazel
564 | | | googletest-break-on-failure-unittest.py
565 | | | googletest-break-on-failure-unittest_.cc
566 | | | googletest-catch-exceptions-test.py
567 | | | googletest-catch-exceptions-test_.cc
568 | | | googletest-color-test.py
569 | | | googletest-color-test_.cc
570 | | | googletest-death-test-test.cc
571 | | | googletest-death-test_ex_test.cc
572 | | | googletest-env-var-test.py
573 | | | googletest-env-var-test_.cc
574 | | | googletest-failfast-unittest.py
575 | | | googletest-failfast-unittest_.cc
576 | | | googletest-filepath-test.cc
577 | | | googletest-filter-unittest.py
578 | | | googletest-filter-unittest_.cc
579 | | | googletest-global-environment-unittest.py
580 | | | googletest-global-environment-unittest_.cc
581 | | | googletest-json-outfiles-test.py
582 | | | googletest-json-output-unittest.py
583 | | | googletest-list-tests-unittest.py
584 | | | googletest-list-tests-unittest_.cc
585 | | | googletest-listener-test.cc
586 | | | googletest-message-test.cc
587 | | | googletest-options-test.cc
588 | | | googletest-output-test-golden-lin.txt
589 | | | googletest-output-test.py
590 | | | googletest-output-test_.cc
591 | | | googletest-param-test-invalid-name1-test.py
592 | | | googletest-param-test-invalid-name1-test_.cc
593 | | | googletest-param-test-invalid-name2-test.py
594 | | | googletest-param-test-invalid-name2-test_.cc
595 | | | googletest-param-test-test.cc
596 | | | googletest-param-test-test.h
597 | | | googletest-param-test2-test.cc
598 | | | googletest-port-test.cc
599 | | | googletest-printers-test.cc
600 | | | googletest-setuptestsuite-test.py
601 | | | googletest-setuptestsuite-test_.cc
602 | | | googletest-shuffle-test.py
603 | | | googletest-shuffle-test_.cc
604 | | | googletest-test-part-test.cc
605 | | | googletest-throw-on-failure-test.py
606 | | | googletest-throw-on-failure-test_.cc
607 | | | googletest-uninitialized-test.py
608 | | | googletest-uninitialized-test_.cc
609 | | | gtest-typed-test2_test.cc
610 | | | gtest-typed-test_test.cc
611 | | | gtest-typed-test_test.h
612 | | | gtest-unittest-api_test.cc
613 | | | gtest_all_test.cc

```

```

614 | | gtest_assert_by_exception_test.cc
615 | | gtest_dirs_test.cc
616 | | gtest_environment_test.cc
617 | | gtest_help_test.py
618 | | gtest_help_test_.cc
619 | | gtest_json_test_utils.py
620 | | gtest_list_output_unittest.py
621 | | gtest_list_output_unittest_.cc
622 | | gtest_main_unittest.cc
623 | | gtest_no_test_unittest.cc
624 | | gtest_pred_impl_unittest.cc
625 | | gtest_premature_exit_test.cc
626 | | gtest_prod_test.cc
627 | | gtest_repeat_test.cc
628 | | gtest_skip_check_output_test.py
629 | | gtest_skip_environment_check_output_test.py
630 | | gtest_skip_in_environment_setup_test.cc
631 | | gtest_skip_test.cc
632 | | gtest_sole_header_test.cc
633 | | gtest_stress_test.cc
634 | | gtest_testbridge_test.py
635 | | gtest_testbridge_test_.cc
636 | | gtest_test_macro_stack_footprint_test.cc
637 | | gtest_test_utils.py
638 | | gtest_throw_on_failure_ex_test.cc
639 | | gtest_unittest.cc
640 | | gtest_xml_outfile1_test_.cc
641 | | gtest_xml_outfile2_test_.cc
642 | | gtest_xml_outfiles_test.py
643 | | gtest_xml_output_unittest.py
644 | | gtest_xml_output_unittest_.cc
645 | | gtest_xml_test_utils.py
646 | | production.cc
647 | | production.h
648 | |
649 | \---googletest-subbuild
650 | | ALL_BUILD.vcxproj
651 | | ALL_BUILD.vcxproj.filters
652 | | CMakeCache.txt
653 | | CMakeLists.txt
654 | | cmake_install.cmake
655 | | googletest-populate.sln
656 | | googletest-populate.vcxproj
657 | | googletest-populate.vcxproj.filters
658 | | ZERO_CHECK.vcxproj
659 | | ZERO_CHECK.vcxproj.filters
660 | |
661 | +---CMakeFiles
662 | | | cmake.check_cache
663 | | | CMakeConfigureLog.yaml
664 | | | generate.stamp
665 | | | generate.stamp.depend
666 | | | generate.stamp.list
667 | | | TargetDirectories.txt

```

```

668 | | |
669 | | | +---1df006e100ab1775d6b8844f8a1a250d
670 | | | | googletest-populate-build.rule
671 | | | | googletest-populate-configure.rule
672 | | | | googletest-populate-download.rule
673 | | | | googletest-populate-install.rule
674 | | | | googletest-populate-mkdir.rule
675 | | | | googletest-populate-patch.rule
676 | | | | googletest-populate-test.rule
677 | | | | googletest-populate-update.rule
678 | | |
679 | | | +---3.29.2
680 | | | | CMakeSystem.cmake
681 | | | | VCTargetsPath.txt
682 | | | | VCTargetsPath.vcxproj
683 | | |
684 | | | +---VCTargetsPath
685 | | | | \---x64
686 | | | | | \---Debug
687 | | | | | | VCTargetsPath.recipe
688 | | | | |
689 | | | | | \---VCTargetsPath.tlog
690 | | | | | VCTargetsPath.lastbuildstate
691 | | | |
692 | | | | \---x64
693 | | | | | \---Debug
694 | | | | +---7e147cec0cdb2f86e12efc8064ce9745
695 | | | | | googletest-populate-complete.rule
696 | | | |
697 | | | | +---ae8f1a22d6213c4e2d2070d31ab9c5d6
698 | | | | | generate.stamp.rule
699 | | | | | googletest-populate.rule
700 | | | |
701 | | | | +---Debug
702 | | | | | googletest-populate-complete
703 | | | |
704 | | | | +---googletest-populate.dir
705 | | | | | Labels.json
706 | | | | | Labels.txt
707 | | | |
708 | | | | \---pkgRedirects
709 | | | +---googletest-populate-prefix
710 | | | | +---src
711 | | | | | v1.14.0.zip
712 | | | | |
713 | | | | | \---googletest-populate-stamp
714 | | | | | | download-googletest-populate.cmake
715 | | | | | | extract-googletest-populate.cmake
716 | | | | | | googletest-populate-patch-info.txt
717 | | | | | | googletest-populate-update-info.txt
718 | | | | | | googletest-populate-urlinfo.txt
719 | | | | | | verify-googletest-populate.cmake
720 | | | | |
721 | | | | | \---Debug

```

```

722 | | | googletest-populate-build
723 | | | googletest-populate-configure
724 | | | googletest-populate-done
725 | | | googletest-populate-download
726 | | | googletest-populate-install
727 | | | googletest-populate-mkdir
728 | | | googletest-populate-patch
729 | | | googletest-populate-test
730 | | | googletest-populate-update
731 | | |
732 | | | \---tmp
733 | | |     googletest-populate-cfgcmd.txt
734 | | |     googletest-populate-mkdirs.cmake
735 | | |
736 | | | \---x64
737 | | |     \---Debug
738 | | |         +---ALL_BUILD
739 | | |         | | ALL_BUILD.recipe
740 | | |         | |
741 | | |         | | \---ALL_BUILD.tlog
742 | | |         | |     ALL_BUILD.lastbuildstate
743 | | |         | |     CustomBuild.command.1.tlog
744 | | |         | |     CustomBuild.read.1.tlog
745 | | |         | |     CustomBuild.write.1.tlog
746 | | |         | |
747 | | |         +---googletest-populate
748 | | |         | |     googletest-populate.recipe
749 | | |         | |
750 | | |         | | \---googlete.DB0C7D59.tlog
751 | | |         | |     CustomBuild.command.1.tlog
752 | | |         | |     CustomBuild.read.1.tlog
753 | | |         | |     CustomBuild.write.1.tlog
754 | | |         | |     googletest-populate.lastbuildstate
755 | | |         | |
756 | | |         \---ZERO_CHECK
757 | | |         |     ZERO_CHECK.recipe
758 | | |         |
759 | | |         \---ZERO_CHECK.tlog
760 | | |             CustomBuild.command.1.tlog
761 | | |             CustomBuild.read.1.tlog
762 | | |             CustomBuild.write.1.tlog
763 | | |             ZERO_CHECK.lastbuildstate
764 | | |
765 | +---src
766 | |     sort.cpp
767 | |     sort.h
768 | |     stringProcessor.cpp
769 | |     stringProcessor.h
770 | |
771 | \---tests
772 |     smoke_test.cpp

```

## Окремо файл build\test\_report.xml:

```
1 <?xml version="1.0" encoding="UTF-8"?>
```

```

2 <testsuites tests="3" failures="0" disabled="0" errors="0" time="0.004"
  ↳ timestamp="2025-11-05T16:03:04.226" name="AllTests">
3   <testsuite name="StringProcessor" tests="2" failures="0" disabled="0" skipped="0"
    ↳ errors="0" time="0.001" timestamp="2025-11-05T16:03:04.227">
4     <testcase name="TrimAndLower"
      ↳ file="C:\Users\artem\Desktop\project\tests\smoke_test.cpp" line="6"
      ↳ status="run" result="completed" time="0."
      ↳ timestamp="2025-11-05T16:03:04.227" classname="StringProcessor" />
5     <testcase name="Palindrome"
      ↳ file="C:\Users\artem\Desktop\project\tests\smoke_test.cpp" line="12"
      ↳ status="run" result="completed" time="0."
      ↳ timestamp="2025-11-05T16:03:04.228" classname="StringProcessor" />
6   </testsuite>
7   <testsuite name="Sort" tests="1" failures="0" disabled="0" skipped="0" errors="0"
    ↳ time="0." timestamp="2025-11-05T16:03:04.229">
8     <testcase name="InsertionSortWorks"
      ↳ file="C:\Users\artem\Desktop\project\tests\smoke_test.cpp" line="19"
      ↳ status="run" result="completed" time="0."
      ↳ timestamp="2025-11-05T16:03:04.229" classname="Sort" />
9   </testsuite>
10 </testsuites>

```

### Ініціалізація локального репозиторію, підключення віддаленого:

```

1 PS C:\Users\artem\Desktop\project> git init
2 Initialized empty Git repository in C:/Users/artem/Desktop/project/.git/
3
4 PS C:\Users\artem\Desktop\project> git add .
5
6 PS C:\Users\artem\Desktop\project> git commit -m "Ініціалізація: CMake + GTest + XML
  ↳ Report"
7 [main (root-commit) 009d10f] Ініціалізація: CMake + GTest + XML Report
8 7 files changed, 174 insertions(+)
9 create mode 100644 .gitignore
10 create mode 100644 CMakeLists.txt
11 create mode 100644 src/sort.cpp
12 create mode 100644 src/sort.h
13 create mode 100644 src/stringProcessor.cpp
14 create mode 100644 src/stringProcessor.h
15 create mode 100644 tests/smoke_test.cpp
16
17 PS C:\Users\artem\Desktop\project> git branch
18 * main
19
20 PS C:\Users\artem\Desktop\project> git remote add origin
  ↳ https://gitea.com/davydchuk/task4lab.git
21
22 PS C:\Users\artem\Desktop\project> git push -u origin main
23 Enumerating objects: 11, done.
24 Counting objects: 100% (11/11), done.
25 Delta compression using up to 16 threads
26 Compressing objects: 100% (10/10), done.
27 Writing objects: 100% (11/11), 2.88 KiB | 589.00 KiB/s, done.
28 Total 11 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)

```

```
29 remote: . Processing 1 references
30 remote: Processed 1 references in total
31 To https://gitea.com/davydchuk/task4lab.git
32 * [new branch]      main -> main
33 branch 'main' set up to track 'origin/main'
```

### Bmict Jenkinsfile:

```
1 pipeline {
2   agent any
3   stages {
4     stage('Checkout') {
5       steps {
6         git branch: 'main',
7           credentialsId: 'gitea-token-id',
8           url: 'https://gitea.com/davydchuk/task4lab.git'
9       }
10    }
11    stage('Build') {
12      steps {
13        bat 'cmake -S . -B build -G "Visual Studio 17 2022"'
14        bat 'cmake --build build --config Debug'
15      }
16    }
17    stage('Test') {
18      steps {
19        bat '.\\build\\Debug\\unit_tests.exe
20          ↪ --gtest_output=xml:build\\test_report.xml'
21      }
22    }
23    post {
24      always {
25        junit 'build/test_report.xml'
26      }
27    }
28  }
```

### Додаємо Jenkinsfile до віддаленого репозиторію:

```
1 PS C:\Users\artem\Desktop\project> git add Jenkinsfile
2 PS C:\Users\artem\Desktop\project> git commit -m "Додавання Jenkinsfile"
3 [main 309c585] Додавання Jenkinsfile
4 1 file changed, 31 insertions(+)
5 create mode 100644 Jenkinsfile
6 PS C:\Users\artem\Desktop\project> git push
7 Enumerating objects: 4, done.
8 Counting objects: 100% (4/4), done.
9 Delta compression using up to 16 threads
10 Compressing objects: 100% (3/3), done.
11 Writing objects: 100% (3/3), 792 bytes | 792.00 KiB/s, done.
12 Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
13 remote: . Processing 1 references
```

```
remote: Processed 1 references in total
To https://gitea.com/davydchuk/task4lab.git
   009d10f..309c585  main -> main
```

### Результат Unit-тестів за допомогою Jenkins:

✔

#4 (Nov 5, 2025, 5:59:25 PM)

🕒

Started by user [Artem Davydochuk](#)

🕒

This run spent:

- 8 ms waiting;
- 25 sec build duration;
- 25 sec total from scheduled to completion.

📦 git

Revision: 309c5859360d132174a19944fda0a8df19caf0f9

Repository: <https://gitea.com/davydchuk/task4lab.git>

- refs/remotes/origin/main

🧪

Tests (no failures)

</>

No changes.

```
[-----] Global test environment set-up.
[-----] 2 tests from StringProcessor
[ RUN      ] StringProcessor.TrimAndLower
[      OK   ] StringProcessor.TrimAndLower (0 ms)
[ RUN      ] StringProcessor.Palindrome
[      OK   ] StringProcessor.Palindrome (0 ms)
[-----] 2 tests from StringProcessor (0 ms total)

[-----] 1 test from Sort
[ RUN      ] Sort.InsertionSortWorks
[      OK   ] Sort.InsertionSortWorks (0 ms)
[-----] 1 test from Sort (0 ms total)

[-----] Global test environment tear-down
[=====] 3 tests from 2 test suites ran. (1 ms total)
[  PASSED  ] 3 tests.

[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Declarative: Post Actions)
[Pipeline] junit
Recording test results
[Checks API] No suitable checks publisher found.
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
```

Tests 3

✔ 3

🕒 Took 0 ms

✎

🎉

All tests are passing!

Nice one! All 3 tests are passing.

All Tests

| Package | Failed | Skipped | Passed | Total | Duration |
|---------|--------|---------|--------|-------|----------|
| (root)  | 0      | 0       | 3 +3   | 3 +3  | 0 ms     |

**Висновок:** У ході виконання лабораторної роботи було розгорнуто систему безперервної інтеграції (Continuous Integration) на основі сервера Jenkins. Було створено віддалений репозиторій у системі керування версіями Gitea та реалізовано повний цикл збірки програмного проєкту з використанням інструментів CMake, GoogleTest і MSBuild. У Jenkins було налаштовано Pipeline із трьома основними етапами:

- **Checkout** — автоматичне отримання коду з віддаленого репозиторію Gitea за допомогою токена доступу (`credentialsId`).
- **Build** — компіляція проєкту за допомогою CMake та генератора Visual Studio 17 2022.

- **Test** — запуск модульних тестів (`unit_tests.exe`) і автоматичне формування звіту у форматі JUnit XML.

Отриманий файл звіту про тести було успішно імпортовано у Jenkins, що дало змогу переглядати результати виконання тестів безпосередньо через веб-інтерфейс. Також було перевірено працездатність збірки та тестів при підключенні репозиторію через SCM (Pipeline script from SCM).

Таким чином, у результаті виконання лабораторної роботи:

1. Ознайомлено з принципами роботи системи безперервної інтеграції Jenkins.
2. Автоматизовано процес збірки, тестування та формування звітів.
3. Реалізовано практичне підключення до системи контролю версій Gitea з використанням токенів доступу.

Робота продемонструвала переваги СІ-підходу у забезпеченні стабільності, відтворюваності та контрольованості процесу розробки програмного забезпечення.